THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 10

Full Syllabus (Chapters 1–13) | Class 8 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** type of flour	The variable being tested
2	**(a)** phases of the Moon	Repeating lunar cycle
3	**(d)** chloroplast	Contains chlorophyll
4	**(d)** protozoan	Spread by mosquitoes
5	**(d)** vector	e.g. mosquito, housefly
6	**(d)** current flows for a longer time	More time gives more heat
7	**(a)** the current is switched off	It is a temporary magnet
8	**(b)** Jupiter	25.4 N
9	**(c)** metal rod that carries charge to the ground	Protects buildings
10	**(a)** high pressure to low pressure	Air moves until pressure equalises
11	**(c)** 100 °C	Liquid to vapour
12	**(d)** metalloids	Intermediate properties
13	**(c)** components are evenly distributed	Same composition throughout
14	**(c)** matter in an object	Measured in kg or g
15	**(b)** Concave mirror – converging	Concave mirror converges
16	**(a)** Chaitra	Starts after the spring equinox
17	**(d)** Karnataka	Where predators and prey were studied
18	**(c)** producer → herbivore → carnivore	Eaten by next level
19	**(a)** Using public transport or cycling	Environment friendly
20	**(d)** endanger life on Earth	Delicate balance

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	Yeast ferments sugar and releases carbon dioxide (1). This makes bread, idli, dosa and cakes soft and spongy (1).	1+1
22	It trains the immune system to recognise germs (1). Immune system then attacks the real pathogen quickly (1).	1+1
23	Amount of current (1). Number of turns of the coil or iron core (1).	1+1
24	Greater height gives greater pressure (1). Water flows with force from taps (1).	1+1
25	Two or more substances mixed without chemical combination (1). Each component keeps its own properties (1).	1+1
26	A uniform mixture of two or more substances (1). Salt in water (1).	1+1
27	Producers (1). Consumers and decomposers (1).	1+1
28	Carbon dioxide (1). Methane (1).	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Yes/no: does the sugar dissolve? (1½). Measurement: time taken in seconds (1½).	1½+1½
30	Formed from lava with trapped gas bubbles (1). Has air pockets so is porous (1). Less dense than water so floats (1).	1+1+1
31	Camphor particles move into air (1). Air particles move freely and mix (1). Particles reach all corners (1).	1+1+1
32	Normal tilts with the mirror (1½). Angle of incidence changes while laws still hold (1½).	1½+1½
33	Months were adjusted to add up to 365 days (1½). February has 28 or 29 days (1½).	1½+1½
34	Decompose waste (1). Fix nitrogen (1). Help in cycles of nutrients (1).	1+1+1
35	Galvani's frog leg twitched with two metals (1). Volta used saltwater paper instead of frog (1). Showed metals and liquid make electricity (1).	1+1+1

	OR: It still contains acids and metals (1). They can cause fires (1). They harm the environment, but can be recycled (1).	1+1+1
36	Reducing frogs raises insects (1). Fewer snakes find food (1). Balance of the food web changes (1).	1+1+1
	OR: Provide conditions to live (1½). Examples: sunlight, water, soil, temperature (1½).	1½+1½

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	Mould on a dish stopped bacteria (1). Penicillin was the first antibiotic (1). Antibiotics treat bacterial infections (1). Vaccines prevent diseases by training immunity (1). Antibiotics do not work on viruses (1).	1+1+1+1+1
	OR: Both build memory (1). Infection involves actual illness (1). Vaccination is safe and controlled (1). Vaccination protects the community (1). Example: chickenpox (1).	1+1+1+1+1
38	Kalboishakhi in West Bengal, Bihar, Jharkhand (1½). Bordoila in Assam (1). Mango showers in southern states (1). Help kharif crops, mangoes and coffee (1½).	1½+1+1+1½
	OR: Less height gives lower pressure (2). P = F/A (1). 900 ÷ 0.3 (1). 3000 N/m² (1).	2+1+1+1
39	Salt–water (1). Alcohol–water (1). Soda water (1). Brass (1). Definition of solution (1).	1+1+1+1+1
	OR: Add solute till no more dissolves (1). Add more solvent (2). Raise temperature (2).	1+2+2
40	National parks (1). Sanctuaries (1). Biosphere reserves (1). Community areas (1). Importance (1).	1+1+1+1+1
	OR: (i) Algae → small fish → big fish (1). (ii) Producer algae, consumers fish (1). (iii) Decomposers recycle (1). (iv) Big fish suffer (1). Algae increase (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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